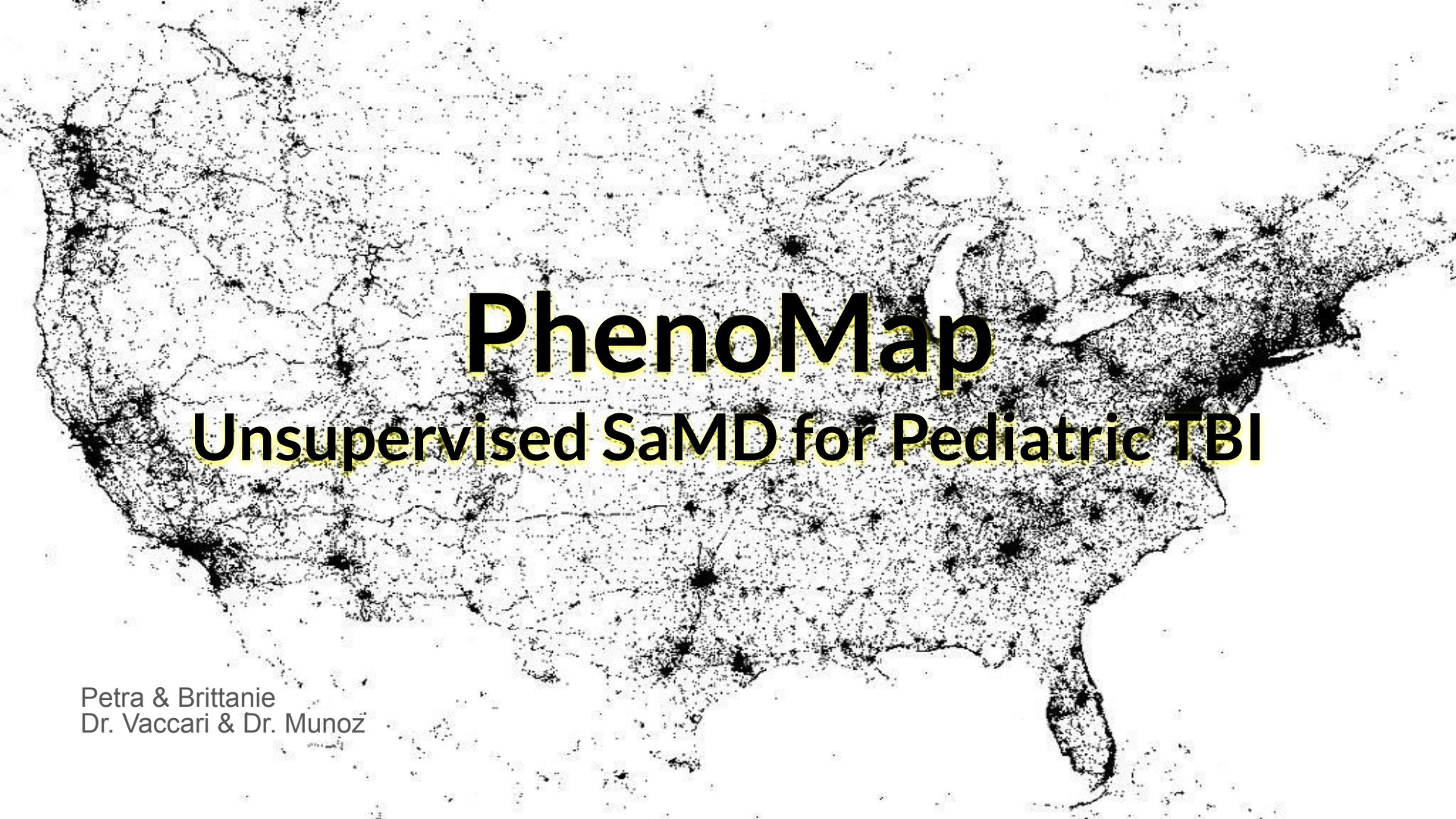




PhenoMap

Unsupervised SaMD for Pediatric TBI

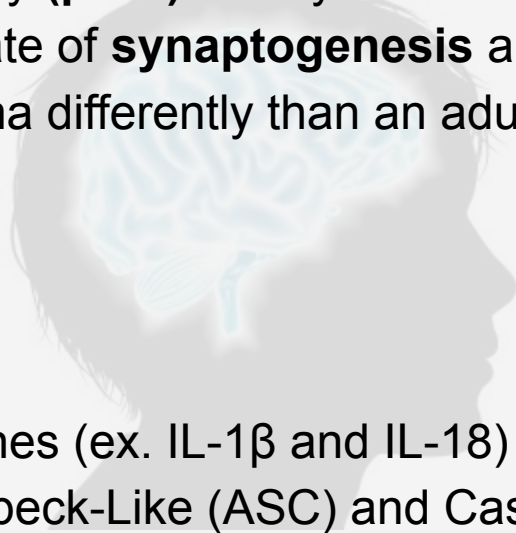
Petra & Brittanie
Dr. Vaccari & Dr. Munoz

Background on pTBI

Pediatric traumatic brain injury (**pTBI**) is a dynamic clinical condition where the young brain, still in a high state of **synaptogenesis** and **developmental plasticity**, responds to trauma differently than an adult brain.

Some Biomarkers:

- Neuroinflammation
- NLRP3 inflammasome
- Pro-inflammatory cytokines (ex. IL-1 β and IL-18)
- Apoptosis-Associated Speck-Like (ASC) and Caspase-1



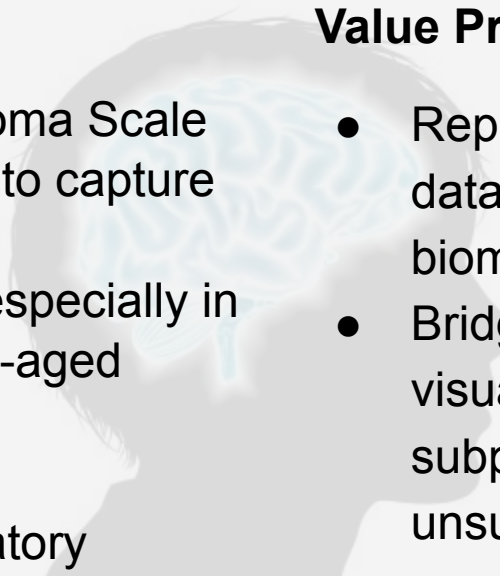
Background on pTBI

Unmet Needs

- Reliance on Glasgow Coma Scale (GCS) and imaging fails to capture the unique inflammatory characteristics in pTBI, especially in non-verbal or pre-school-aged children
- No existing tools identify patient-specific inflammatory trajectories

Value Proposition

- Replaces the subjective GCS with data-driven blood-based biomarkers
- Bridges the gap in injury visualization by discovering hidden subphenotypes through unsupervised clustering

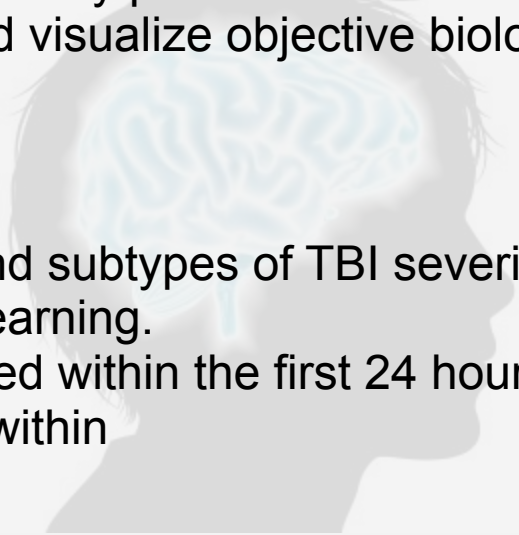


PhenoMap Introduction

The software's intended for use by pediatric intensive care clinicians and neurosurgeons to identify and visualize objective biological subphenotypes in pediatric patients.

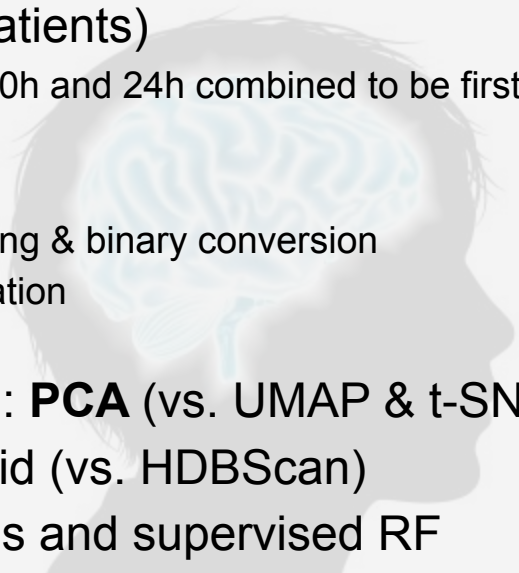
Goal:

- Discover new classes and subtypes of TBI severity that moves beyond the GCS via unsupervised learning.
- It is designed to be utilized within the first 24 hours following a traumatic brain injury (TBI), exclusively within
 - tertiary trauma centers,
 - emergency rooms, and
 - pediatric intensive care units (PICUs)

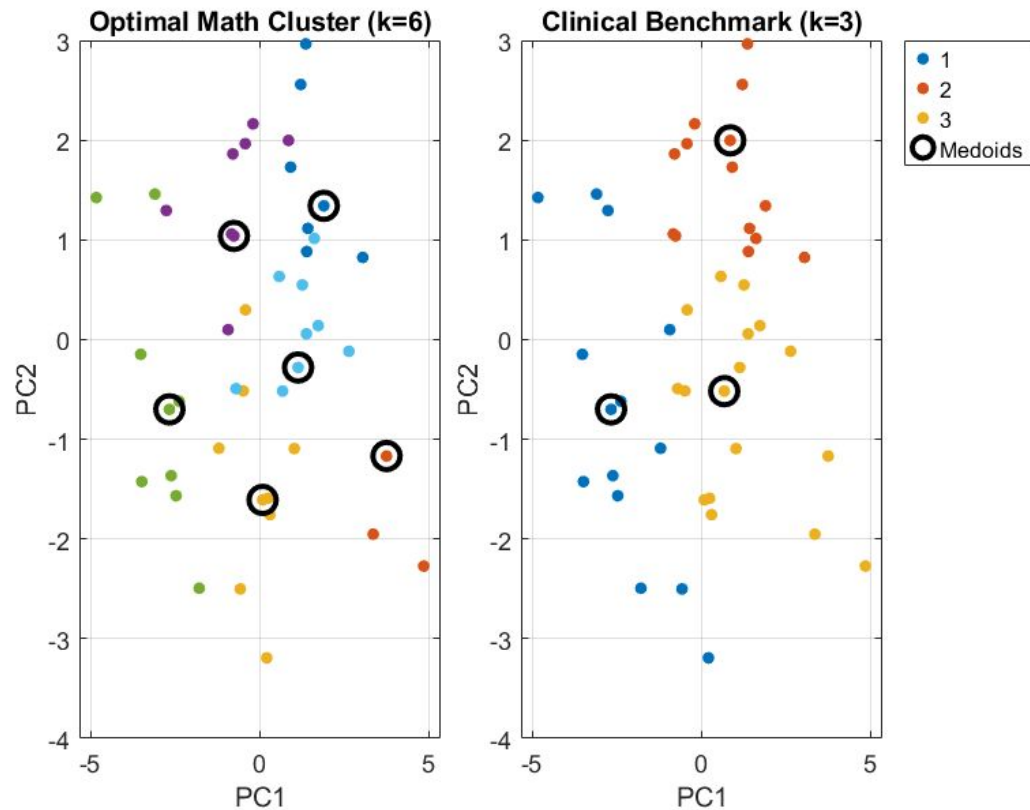


Our Approach

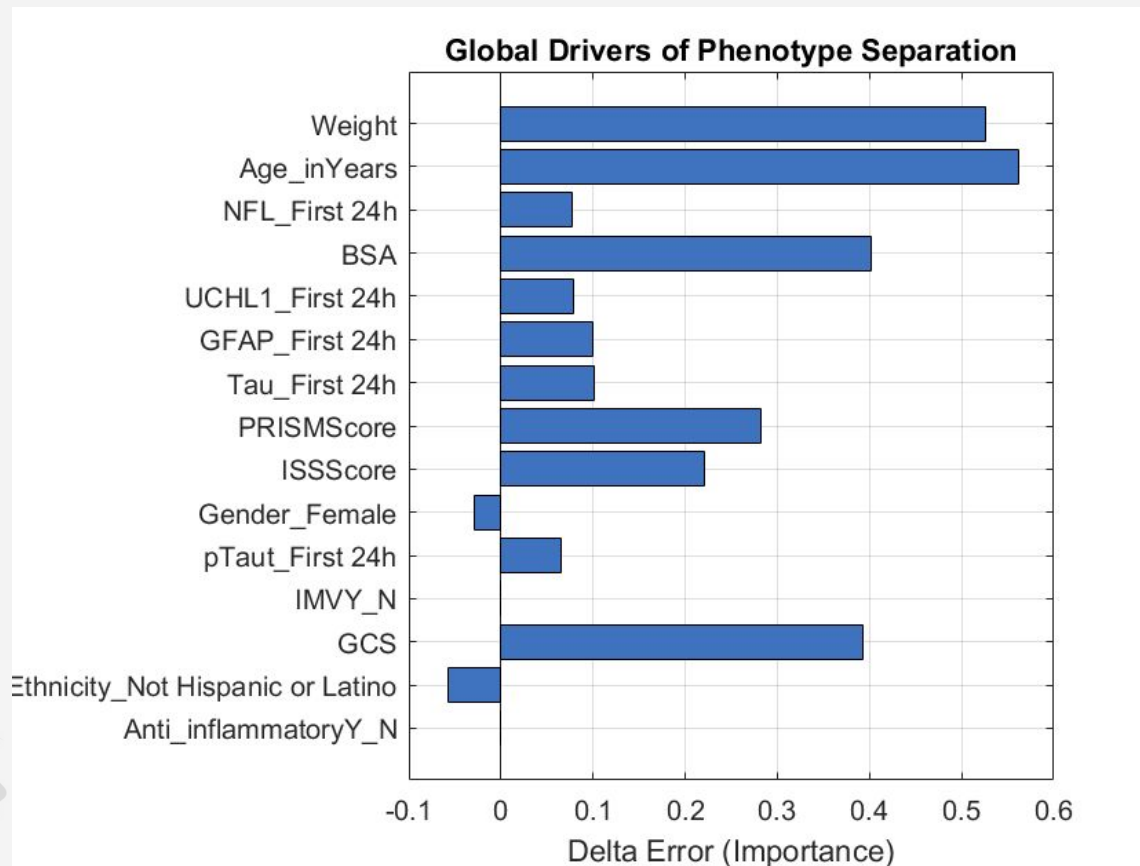
- Preprocessing: (for 51 patients)
 - Features: all biomarkers (0h and 24h combined to be first 24h), GCS scores, weight, age (in years)
 - Outliers: log scale
 - Encoding: one-hot encoding & binary conversion
 - Missing data: KNN imputation
 - Z-score normalization
- Dimensionality reduction: **PCA** (vs. UMAP & t-SNE for additional visuals)
- Clustering: **PAM**/k-medoid (vs. HDBScan)
- Validation: Kruskal-Wallis and supervised RF



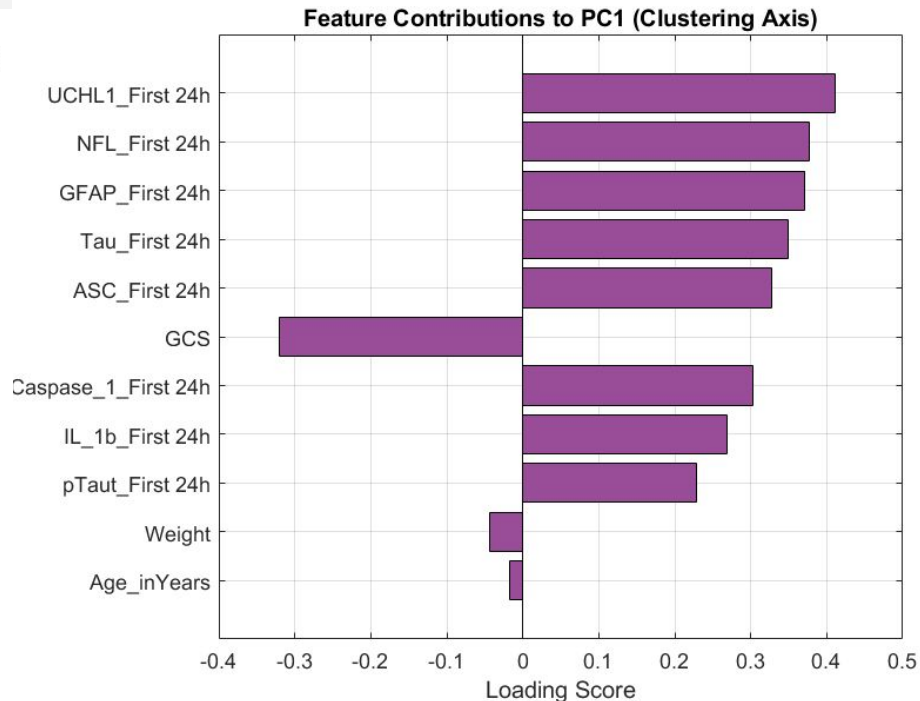
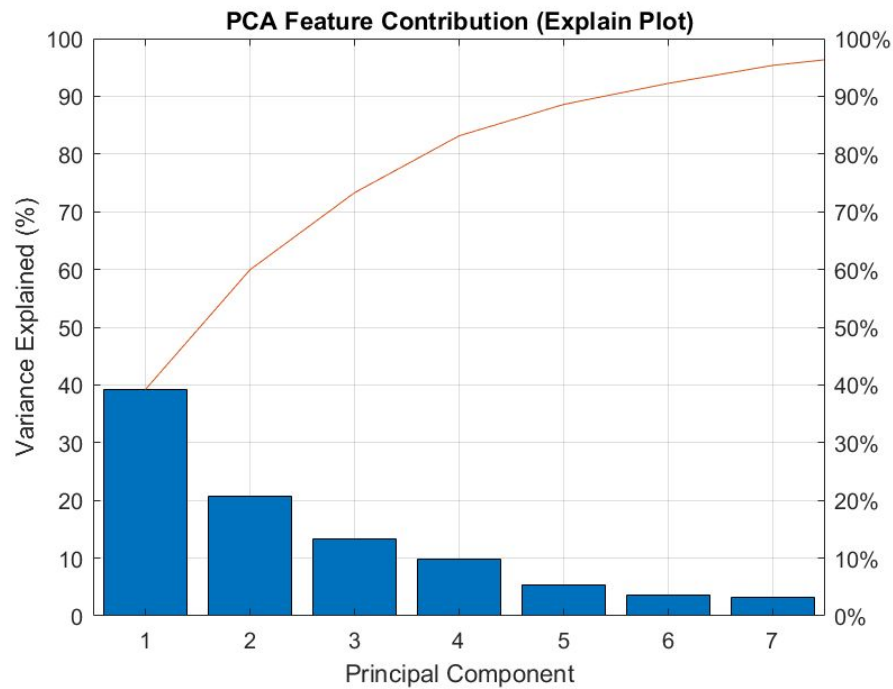
Human-in-the-Loop



Interpretability (Global Drivers)

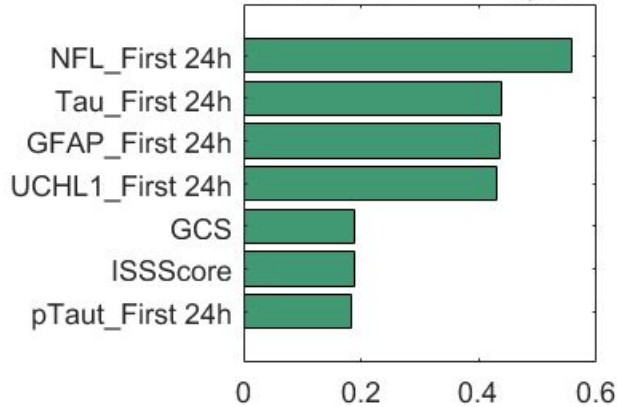


Interpretability (PCA Explain & Loading Plots)

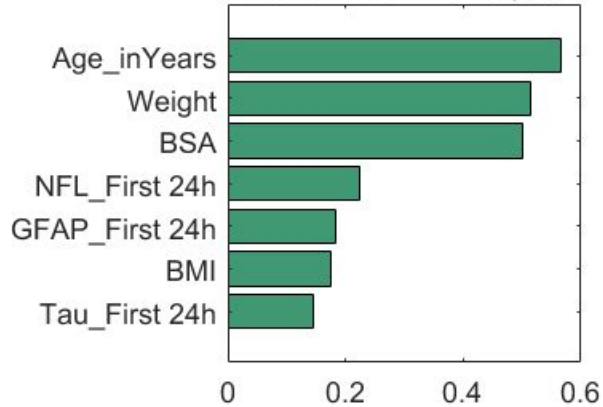


Interpretability (Local Drivers: One-vs-Rest)

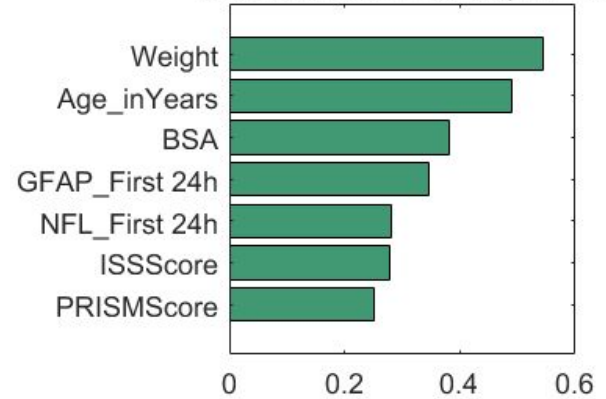
What defines Phenotype 1?



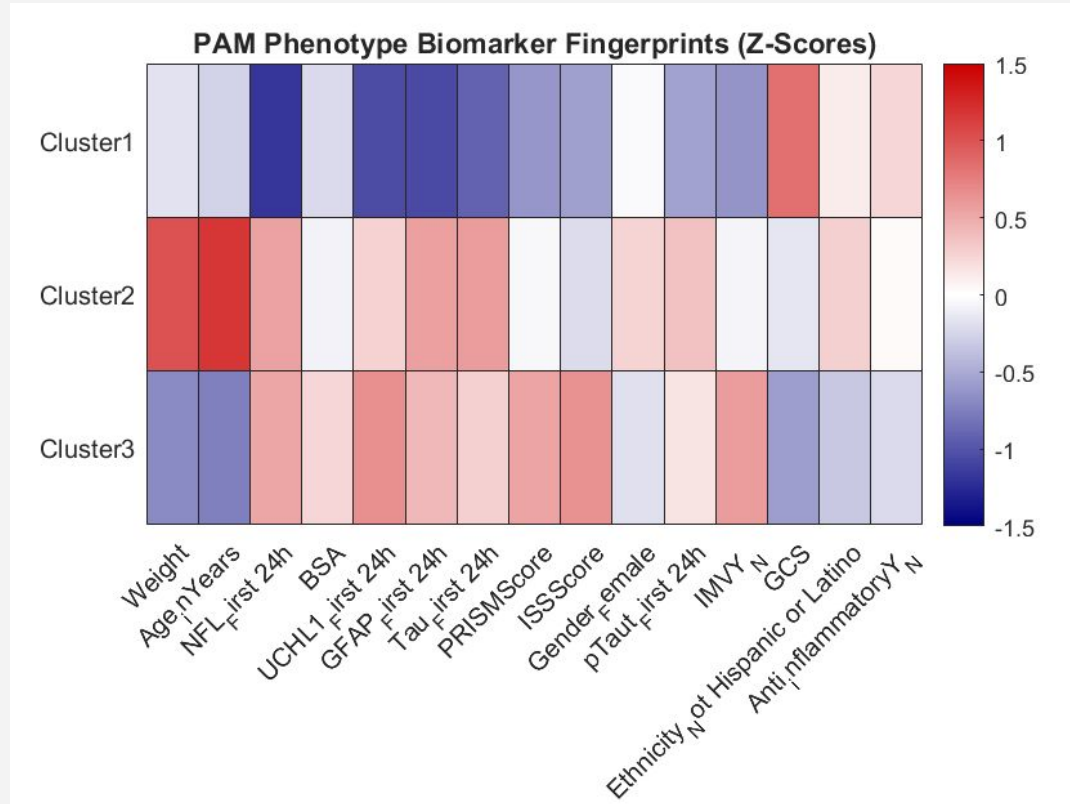
What defines Phenotype 2?



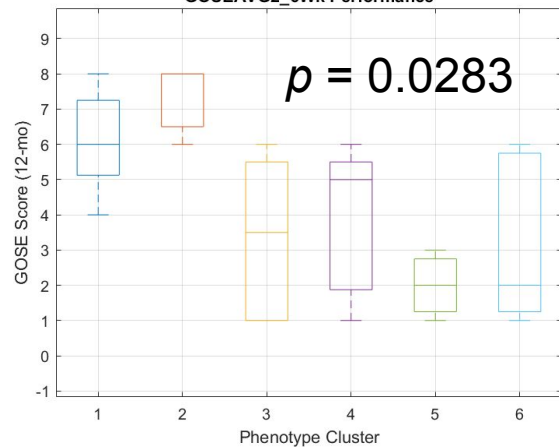
What defines Phenotype 3?



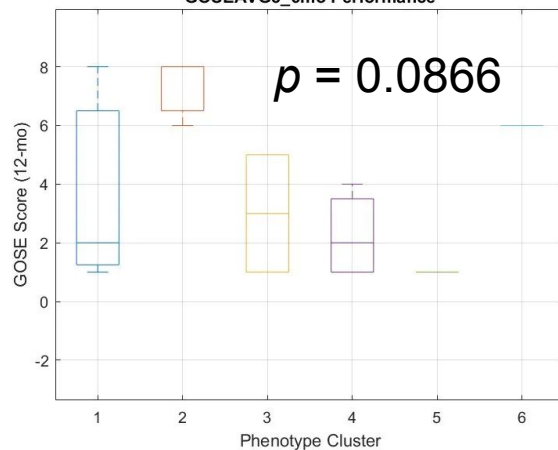
Interpretability



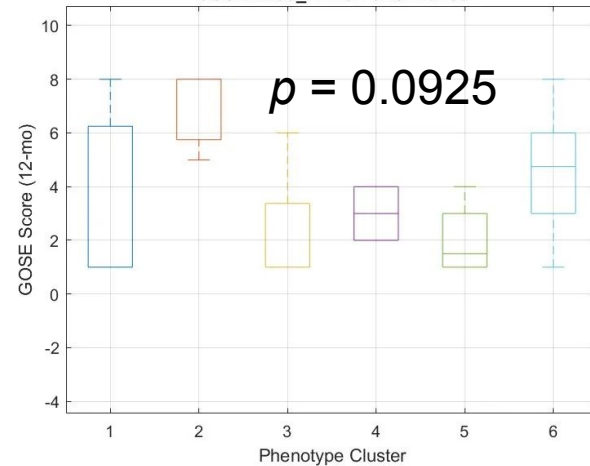
Optimal Math Cluster (k=6)
GOSEAVG2_6Wk Performance



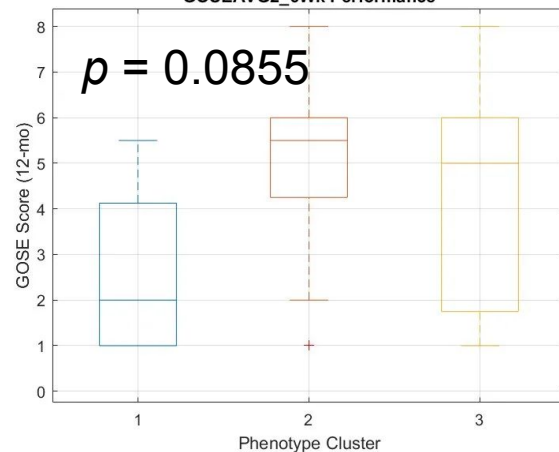
Optimal Math Cluster (k=6)
GOSEAVG3_6mo Performance



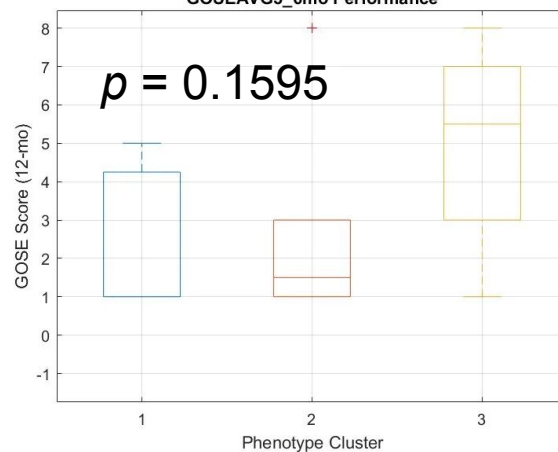
Optimal Math Cluster (k=6)
GOSEAVG9_12mo Performance



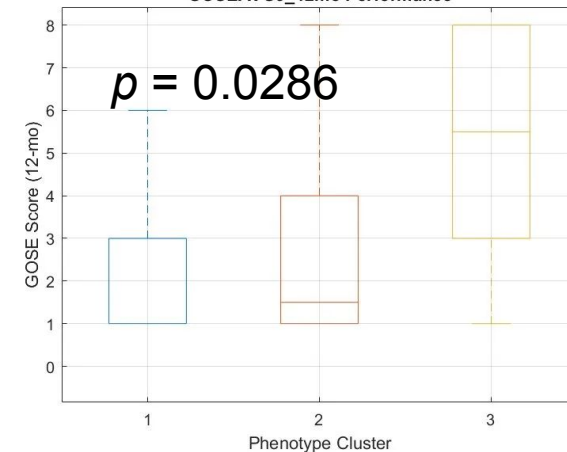
Clinical Benchmark (k=3)
GOSEAVG2_6Wk Performance



Clinical Benchmark (k=3)
GOSEAVG3_6mo Performance



Clinical Benchmark (k=3)
GOSEAVG9_12mo Performance



Supervised-based Validation

Random Forest
OOB error: 0.089
(accuracy: 91.1%)



Hazard Traceability Matrix

1. Non-representative training dataset (bias in pediatric data); Po/S = 3/5
 - a. model learns skewed relationships which leads to poor generalization
2. Missing or incomplete biomarker inputs (ASC, Caspase-1 at 0h/24h); Po/S = 4,4
 - a. leads to imputation or exclusion; Delayed or inappropriate treatment decisions
3. Incorrect preprocessing (normalization, encoding errors); Po/S = 2/4
 - a. Improper scaling or encoding -> altered feature distributions -> clustering distortion and incorrect phenotype grouping
4. Limited interpretability of clustering outputs (UMAP + SHAP); Po/S = 3/4
5. Cluster instability across datasets (lack of reproducibility); Po/S = 3/4
6. Overfitting to training data; Po/S = 3/5
7. Overreliance on SaMD output by clinicians; Po/S = 2/5

Requirement Traceability Matrix

GMLP:

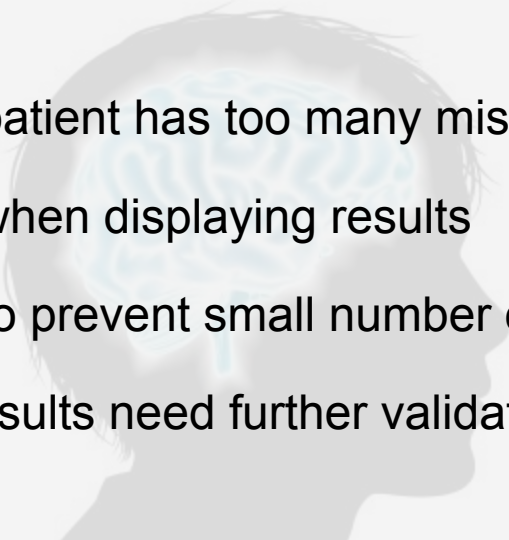
1. Shows which features drives cluster/phenotype separation
2. Allows clinician to view individual patient data by selecting a data point

GUI:

3. Displays scatter plot with color-coded phenotype cluster assignments
4. Shows side-by-side comparison of algorithm output and per phenotype biomarker heatmap
5. Display GOS-E outcome comparisons at 6 weeks, 6 months, and 12 months

Requirement Traceability Matrix

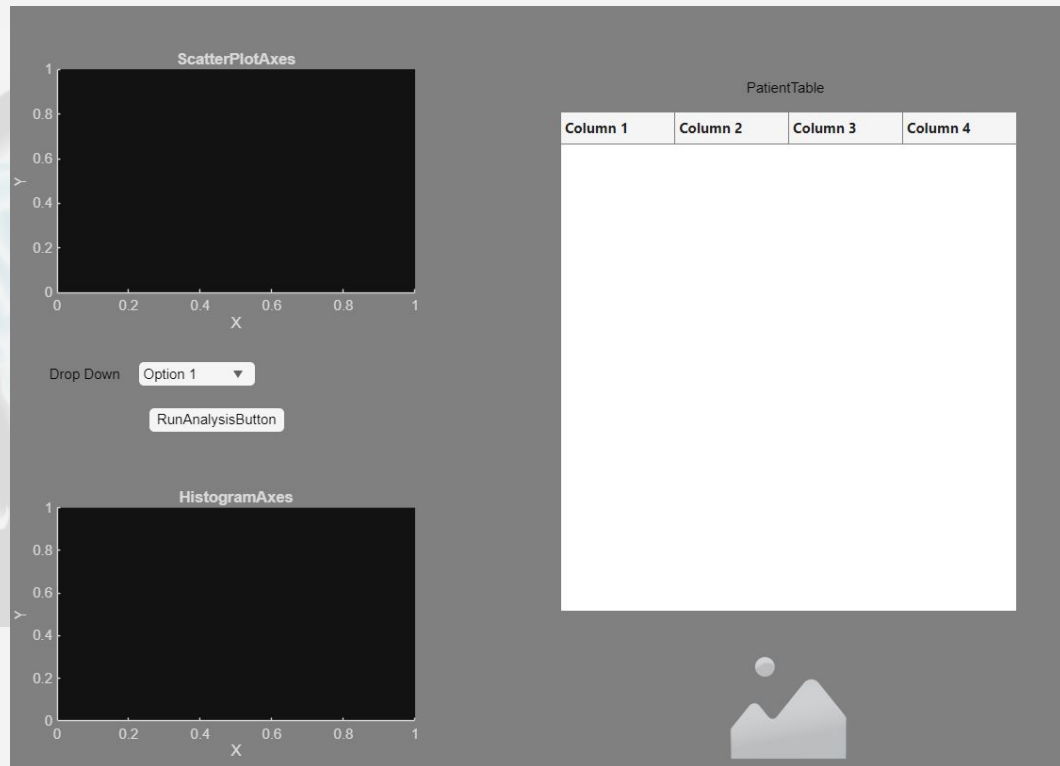
Hazard:

- 
6. Warns clinician when a patient has too many missing biomarker data
 7. Shows model reliability when displaying results
 8. Locks clustering to $k=3$ to prevent small number of patients per cluster
 9. Reminds clinician that results need further validation and human-in-the-loop, not a prognosis
 10. Alerts clinician when patient biomarker values are outside of range

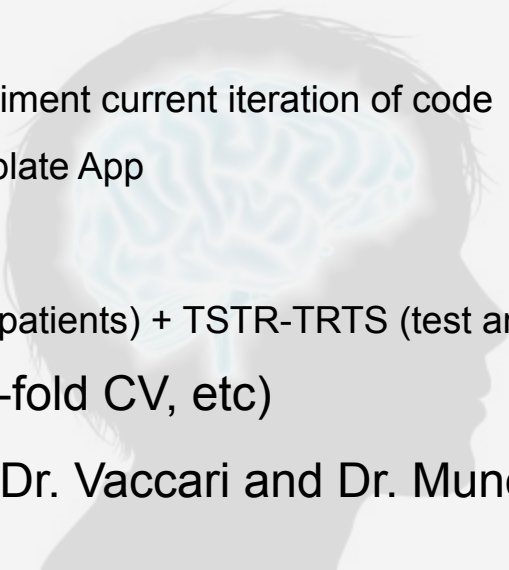
App Progress

Goals:

- t-SNE / UMAP plot (when selecting a dot, it shows the patient's data; alert DBScan outliers)
- Histogram & median of GOSE scores of each clusters (to further validate whether the clusters were correct)
- Raw data (age, weight, etc)



Left To Do

- App Refinement
 - Adjust Callbacks and Impliment current iteration of code
 - Possibly fall back on Template App
 - Augmentation
 - SMOTE (create synthetic patients) + TSTR-TRTS (test and train on synthetic and real patients)
 - Continued Validation (10-fold CV, etc)
 - Interpret the results with Dr. Vaccari and Dr. Munoz
- 

Feedback?

